

# FeCode\_updated2025\_20260225

2026-02-27

## Things that need to be changed

```
Date_Run = "20260225"
plates<- c("Plate1", "Plate2", "Plate3")
Std_plates<- c("STD")
Run_by = "Zoe Read"#Instrument user
Script_run_by = "Zoe Read"#Code user
Project = "COMPASS_Synoptic_CB"
Experiment="2025"
Run_notes="The check stds for FE3 60 uM were all bad. Spikes were bad. "#any notes from run

#Stds that should be excluded
stds_to_remove<-c("")
```

## File paths

```
## [1] "Tidy Data/20260225_COMPASS_Synoptic_CB_Fe_STD.csv"

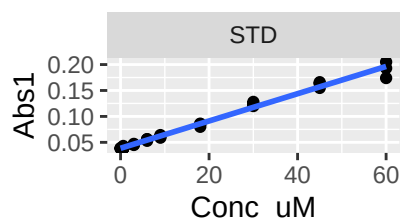
## [1] "Tidy Data/20260225_COMPASS_SynopticCB_Fe_Plate1.csv"
## [2] "Tidy Data/20260225_COMPASS_SynopticCB_Fe_Plate2.csv"
## [3] "Tidy Data/20260225_COMPASS_SynopticCB_Fe_Plate3.csv"
```

## Plot standards

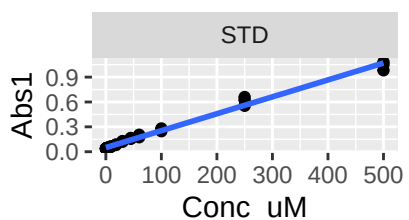
Table 1: Slopes

| Date     | Project                       | Curve | R2        | Slope     | Intercept | Top_STD |
|----------|-------------------------------|-------|-----------|-----------|-----------|---------|
| 20260225 | COMPASS_Synoptic_CFE_2_low    |       | 0.9892313 | 0.0026299 | 0.0386026 | 60      |
| 20260225 | COMPASS_Synoptic_CFE_2_high   |       | 0.9908838 | 0.0020407 | 0.0499991 | 500     |
| 20260225 | COMPASS_Synoptic_CFE_Tot_low  |       | 0.9875164 | 0.0015717 | 0.0411775 | 60      |
| 20260225 | COMPASS_Synoptic_CFE_Tot_high |       | 0.9936050 | 0.0015964 | 0.0409286 | 100     |
| 20260225 | COMPASS_Synoptic_CFE_2_abs2   |       | 0.9936974 | 0.0040726 | 0.0424826 | 100     |
| 20260225 | COMPASS_Synoptic_CFE_3_abs1   |       | 0.5159600 | 0.0000091 | 0.0389755 | 500     |
| 20260225 | COMPASS_Synoptic_CFE_2_diff   |       | 0.9931472 | 0.0016693 | 0.0014384 | 100     |

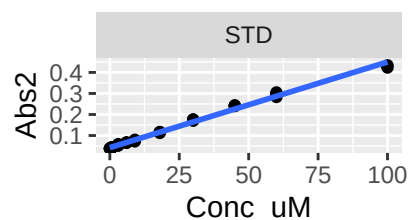
Low Fe(II) Absorbance



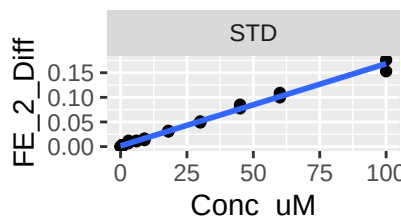
High Fe(II) Absorbance



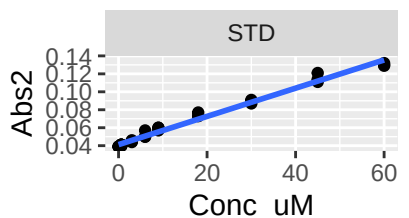
Fe(II) Absorbance 2



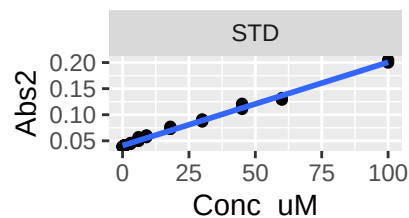
Fe(II) Absorbance D



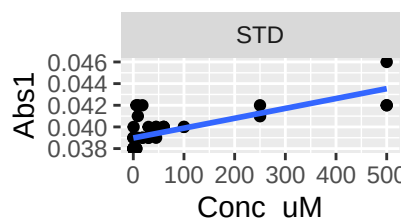
Low Fe(III) Absorbance



High Fe(III) Absorbance

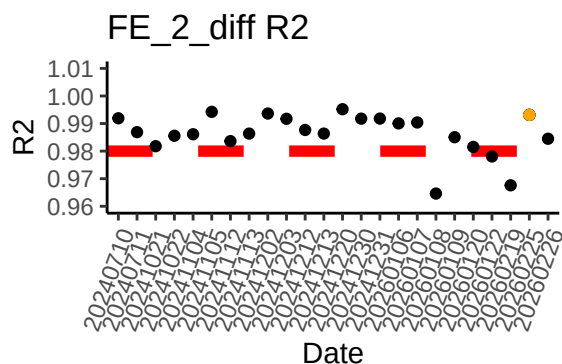
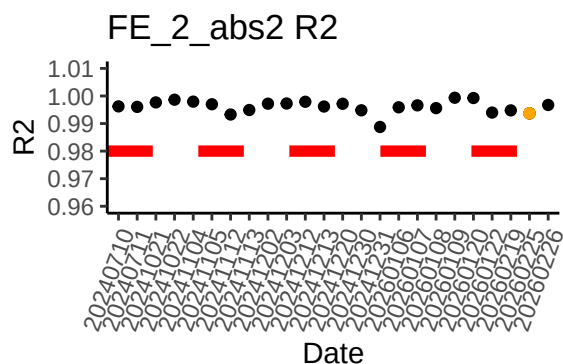
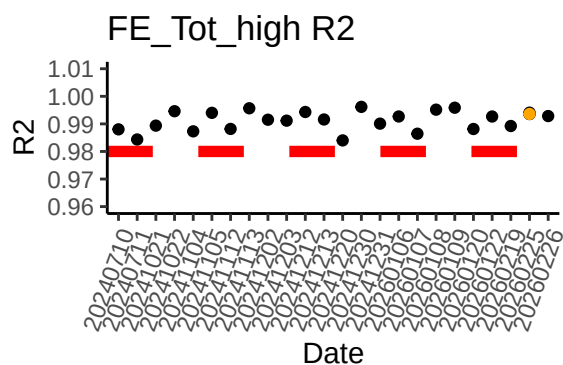
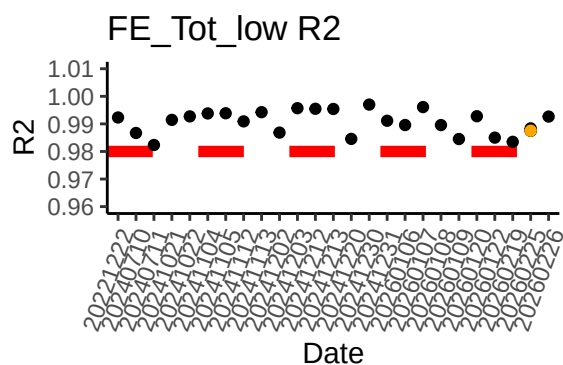
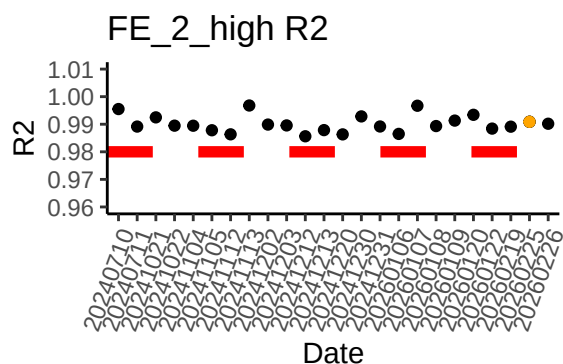
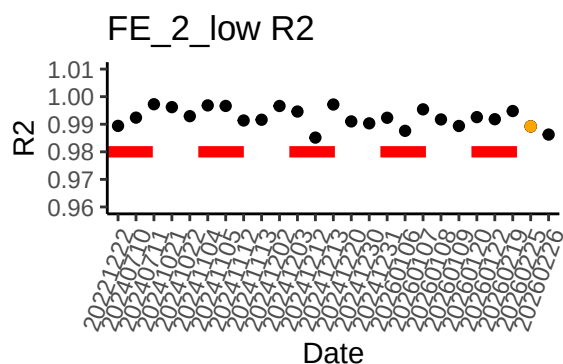


Fe(III) Absorbance 1 Curve



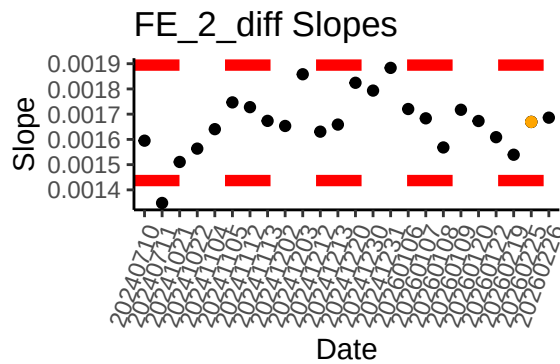
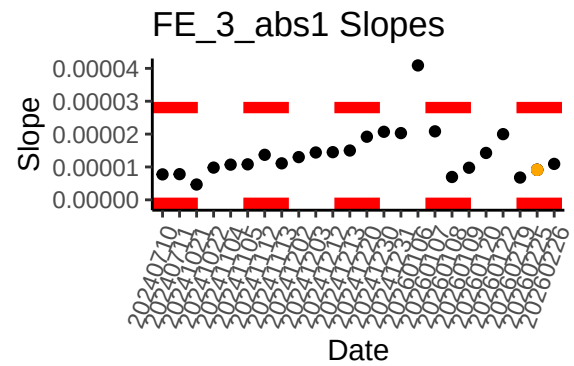
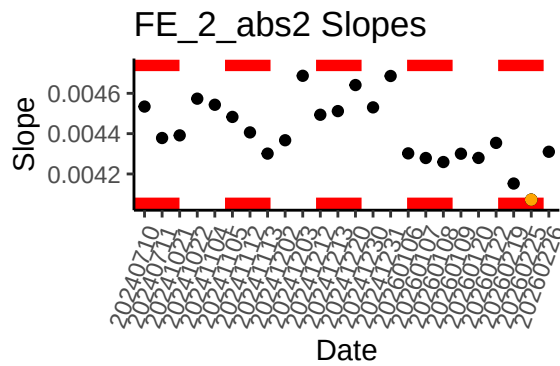
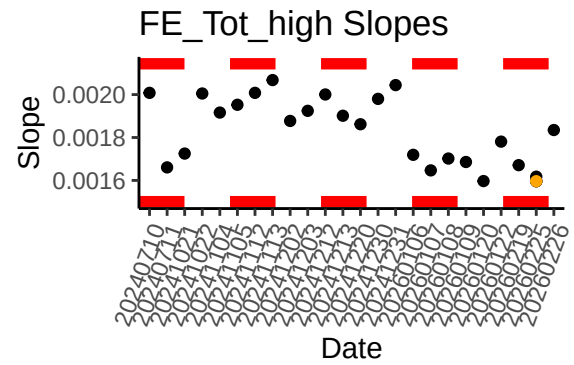
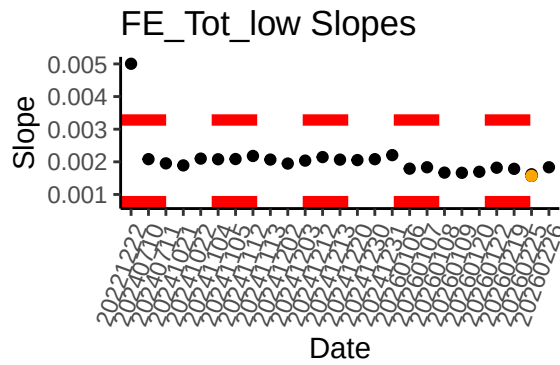
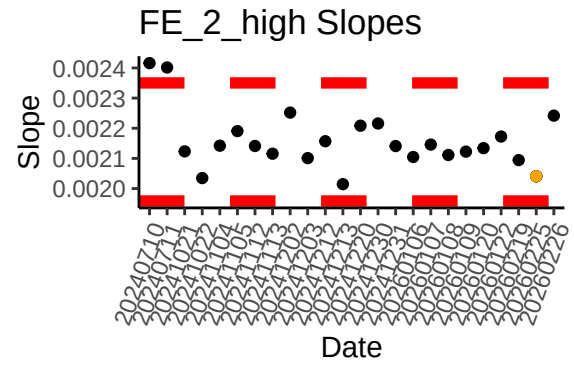
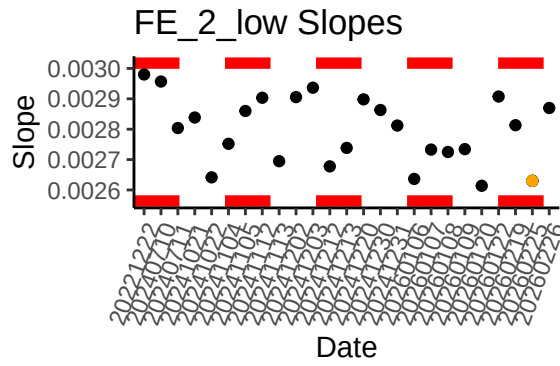
## Checking STD Data against QAQC file

R2



```
## [1] "FE_2_low: Std Curve r2 GOOD"
## [1] "FE_2_high: Std Curve r2 GOOD"
## [1] "FE_Tot_low: Std Curve r2 GOOD"
## [1] "FE_Tot_high: Std Curve r2 GOOD"
## [1] "FE_2_abs2: Std Curve r2 GOOD"
## [1] "FE_3_abs1: R2 will always be low"
## [1] "FE_2_diff: Std Curve r2 GOOD"
```

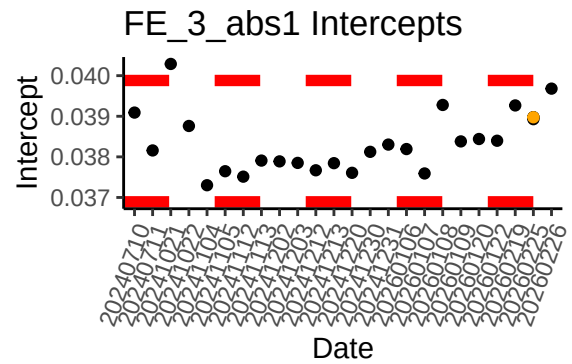
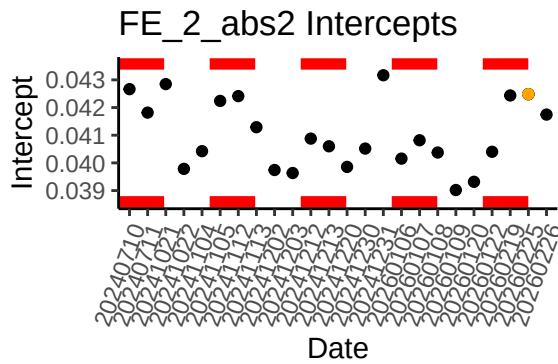
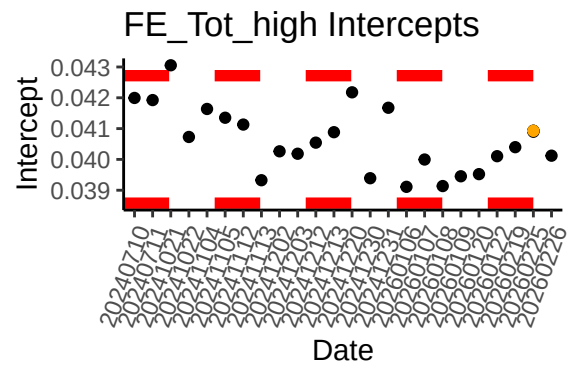
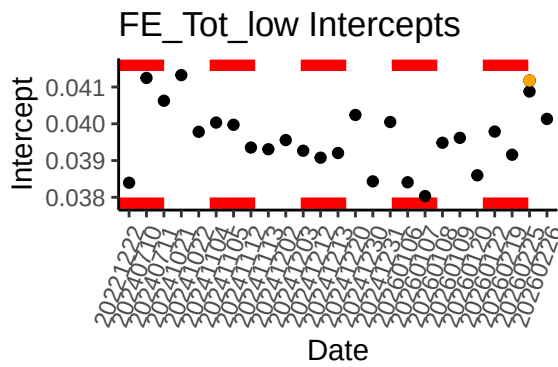
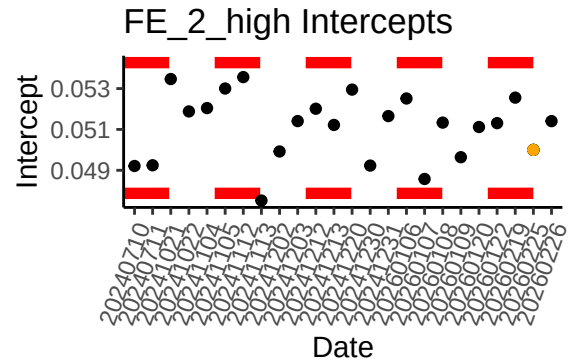
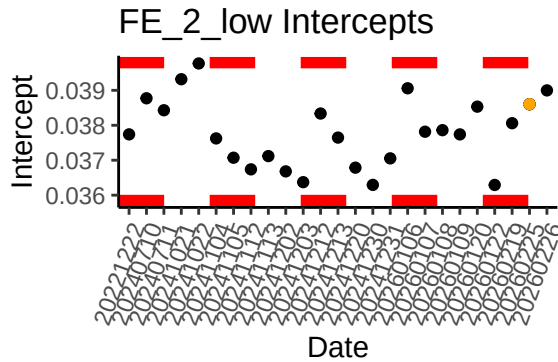
## Slopes

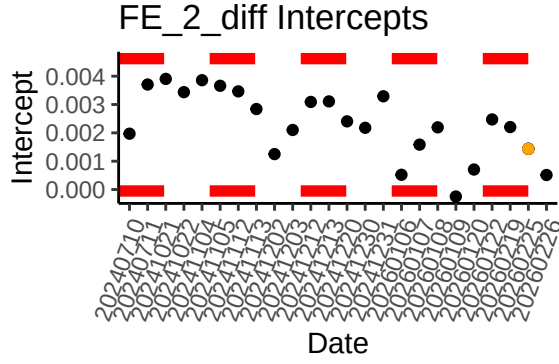


```
## [1] "FE_2_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_high:Std curve slope is with 2 sd of previous slopes"
```

```
## [1] "FE_Tot_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_Tot_high:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_abs2:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_3_abs1:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_diff:Std curve slope is with 2 sd of previous slopes"
```

## Intercepts



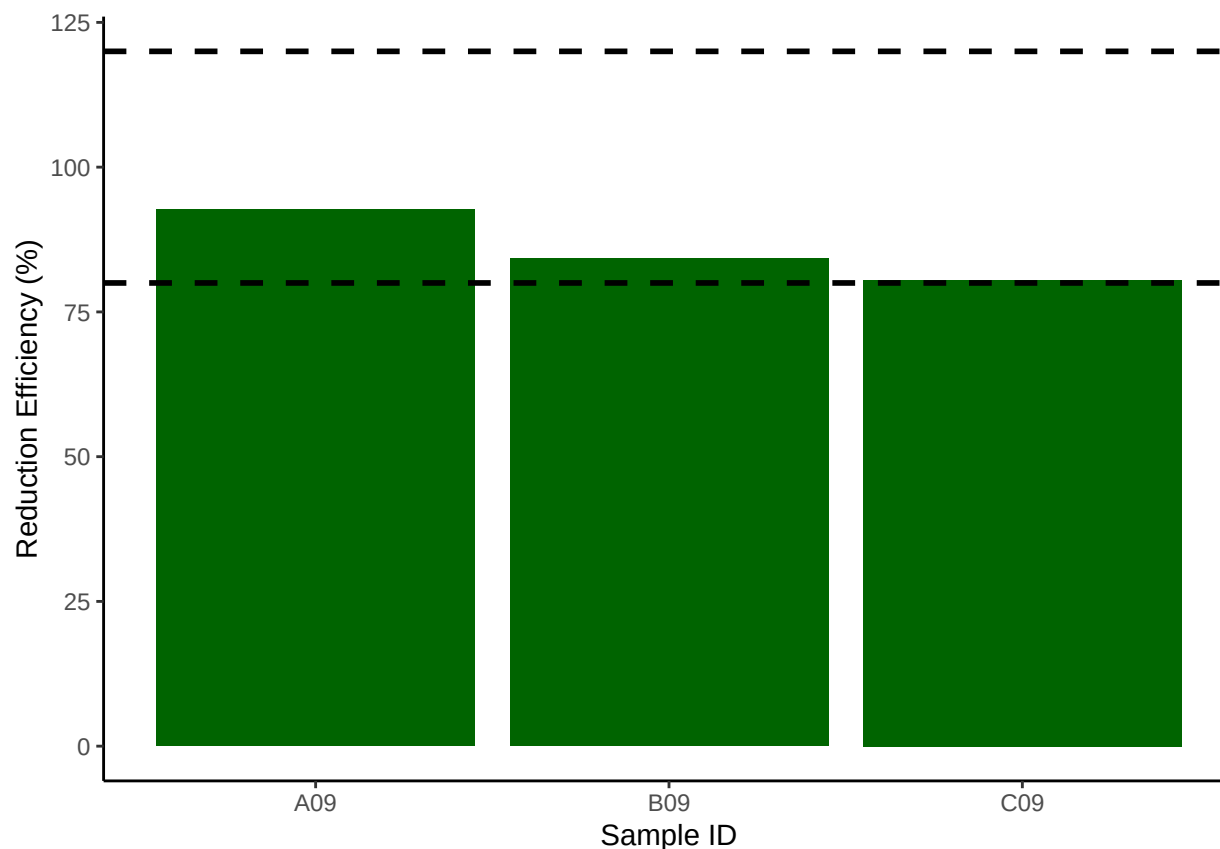


```
## [1] "FE_2_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_abs2:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_3_abs1:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_diff:Std curve intercept is with 2 sd of previous intercepts"
```

## Calculate Reduction Efficiency

Table 2: Reduction Efficiency

| ID  | FE2   | FE3   | Eff      | Eff1     | Eff_flag |
|-----|-------|-------|----------|----------|----------|
| A09 | 0.174 | 0.129 | 74.13793 | 92.67241 | OK       |
| B09 | 0.193 | 0.130 | 67.35751 | 84.19689 | OK       |
| C09 | 0.205 | 0.132 | 64.39024 | 80.48780 | OK       |



```
## [1] ">60% of Reduction Efficiencies are within range"
```

## Method Minimum Detection Limit (MDL) Calculation

I use the lowest standard to calculate MDL How to calculate MDL [https://www.epa.gov/sites/default/files/2016-12/documents/mdl-procedure\\_rev2\\_12-13-2016.pdf](https://www.epa.gov/sites/default/files/2016-12/documents/mdl-procedure_rev2_12-13-2016.pdf) Look at Table 1 on page 5 The Student's t-value used to calculate the method detection limit (MDL) is the one appropriate for a 99% confidence level and a standard deviation estimate with n-1 degrees of freedom

```
## [1] 0.003333231
```

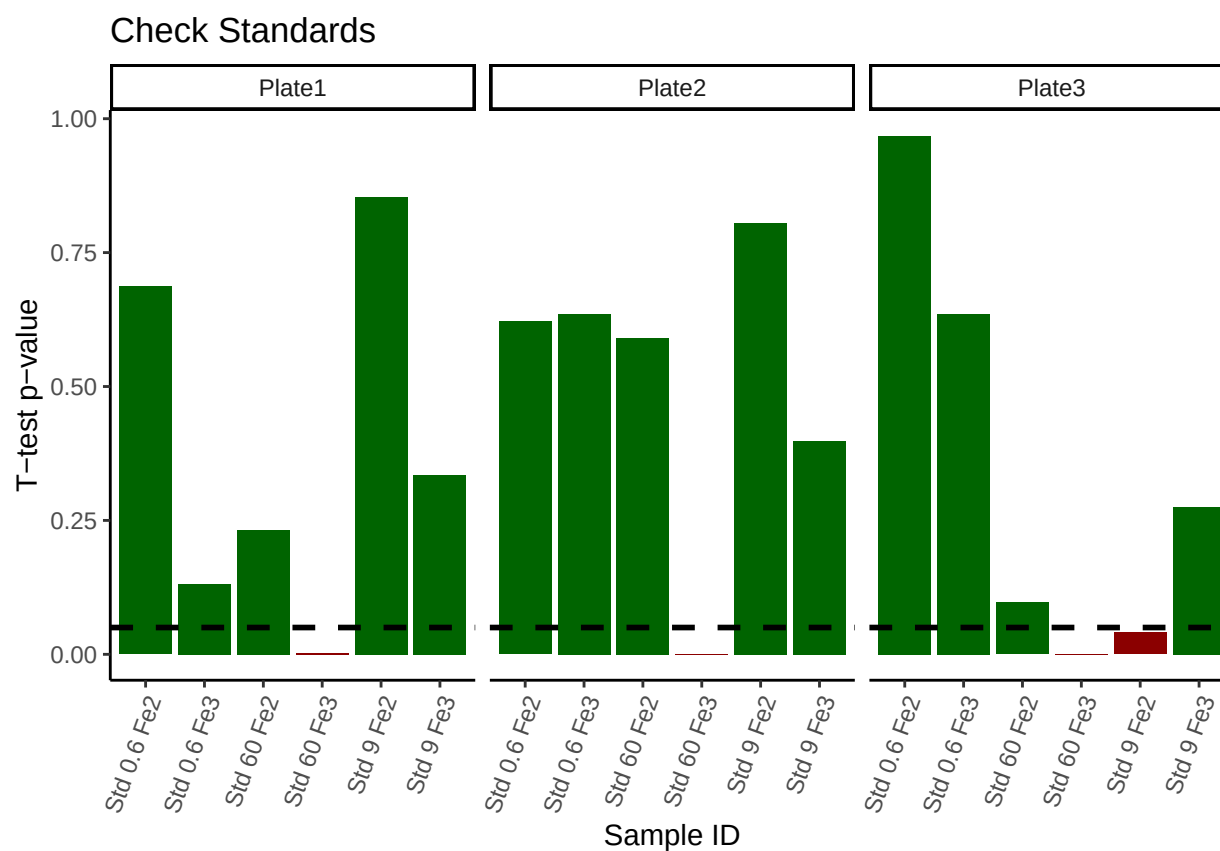
## Check standards QAQC

```
## [1] "Plate1"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate2"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
```

```

## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate3"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"

```

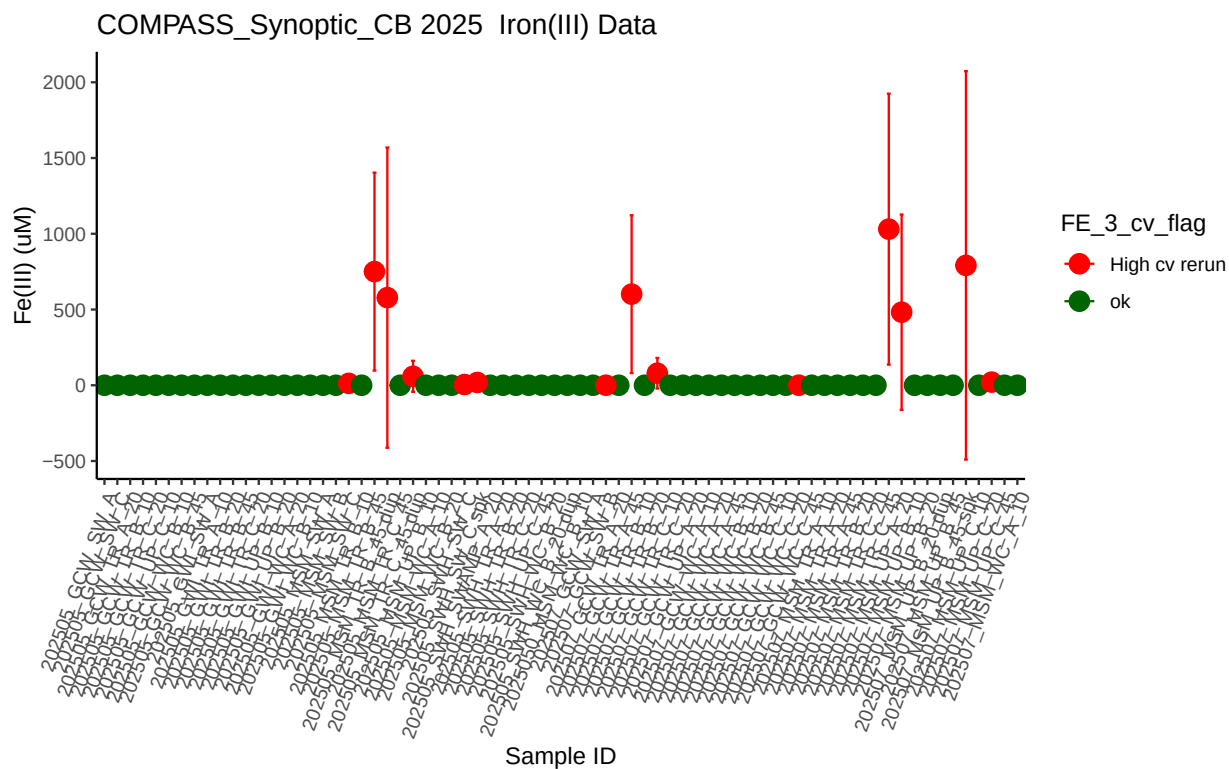
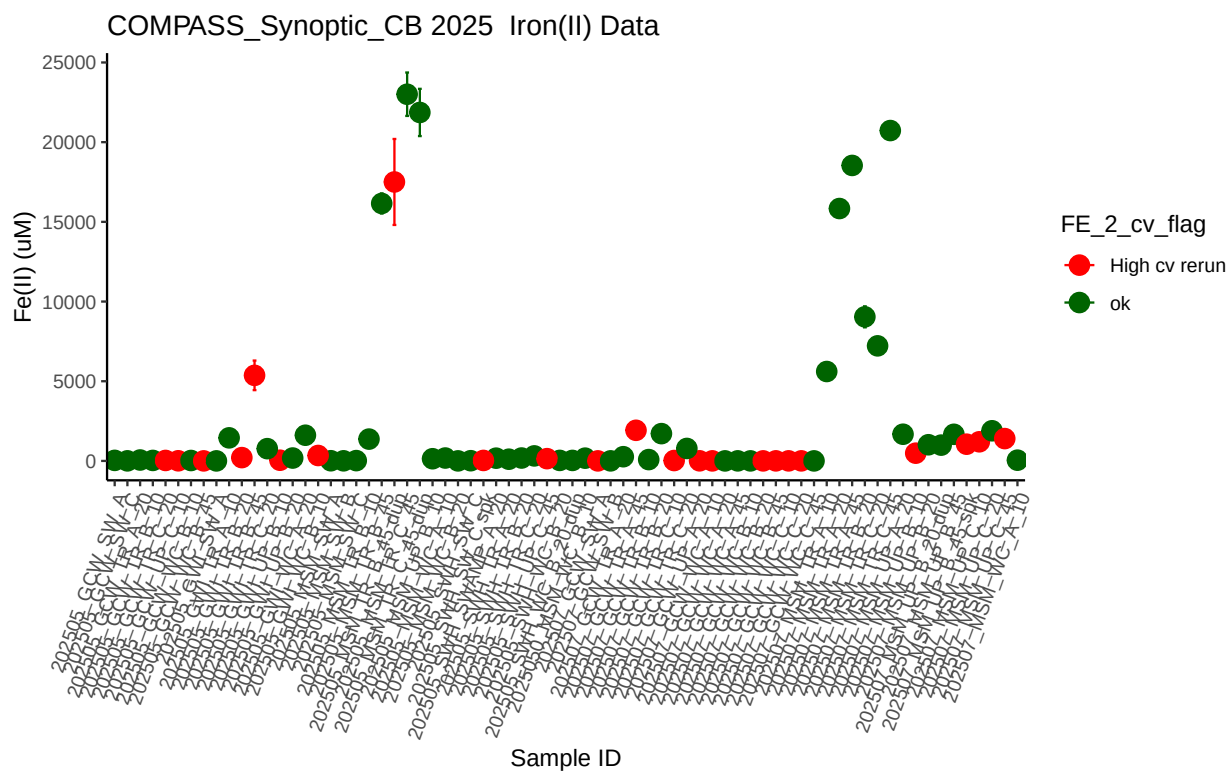


#No Matrix Checks We do not use matrix checks for the Fe samples because our matrix checks are not made with Trace metal grade reagents. If we want to do this in the future, we need to use trace metal grade sodium chloride and sodium bicarbonate for making the matrix checks



## Subset Data and Calculate Concentrations

Calculate Averages across wells, std. dev, and cv.



# Remove Bad Reps

## Flagged data

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##     group_rows
```

Table 3: High CV Samples

| ID full                          | FE 2 mean    | FE 2 cv   | FE 3 mean   | FE 3 cv   | FE 2 cv flag  | FE 3 cv flag  |
|----------------------------------|--------------|-----------|-------------|-----------|---------------|---------------|
| Plate1_1_202505_MSM_SW_C         | 20.050244    | 6.096162  | 7.151613    | 19.76915  | ok            | High cv rerun |
| Plate1_50_202505_MSM_TR_B_45 dup | 15950.492820 | 1.955131  | 867.621972  | 139.34219 | ok            | High cv rerun |
| Plate2_1_202505_SWH_SW_C         | 13.712938    | 4.235589  | 3.899869    | 14.72999  | ok            | High cv rerun |
| Plate2_1_202505_SWH_SW_C spk     | 34.562675    | 8.557094  | 9.996200    | 22.20258  | ok            | High cv rerun |
| Plate2_8_202507_GCW_TR_B_20      | 1713.147852  | 9.141849  | 119.374601  | 87.16395  | ok            | High cv rerun |
| Plate3_8_202507_MSM_UP_A_20      | 1675.252215  | 6.503313  | 111.819475  | 82.31237  | ok            | High cv rerun |
| Plate3_8_202507_MSM_UP_B_45 spk  | 903.618641   | 7.055700  | 1187.006564 | 129.04844 | ok            | High cv rerun |
| Plate1_8_202505_GWI_UP_B_10      | 40.753610    | 10.555890 | 0.000000    | 0.00000   | High cv rerun | ok            |
| Plate2_1_202507_GCW_UP_A_20      | 5.854678     | 18.369525 | 0.000000    | 0.00000   | High cv rerun | ok            |
| Plate2_8_202505_GWI_TR_B_45      | 4888.540885  | 11.794555 | 0.000000    | 0.00000   | High cv rerun | ok            |
| Plate3_1_202507_GCW_WC_C_10      | 4.333724     | 12.408224 | 0.000000    | 0.00000   | High cv rerun | ok            |

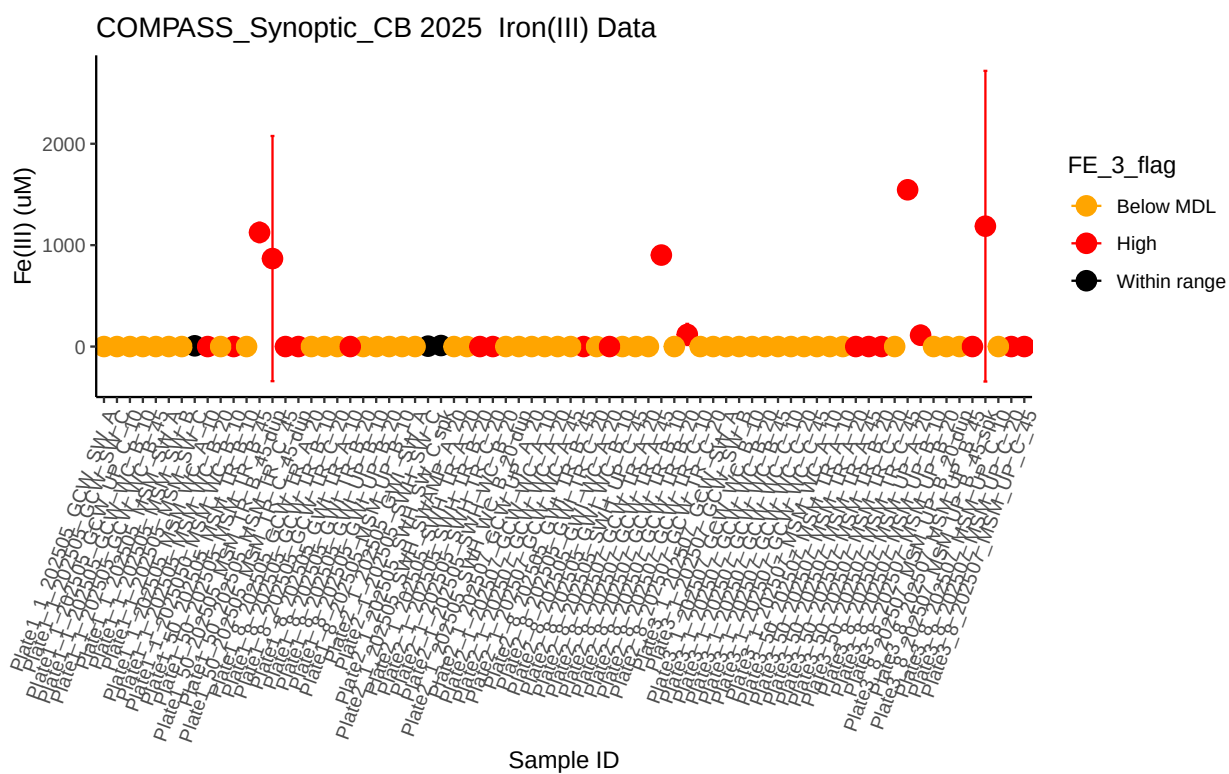
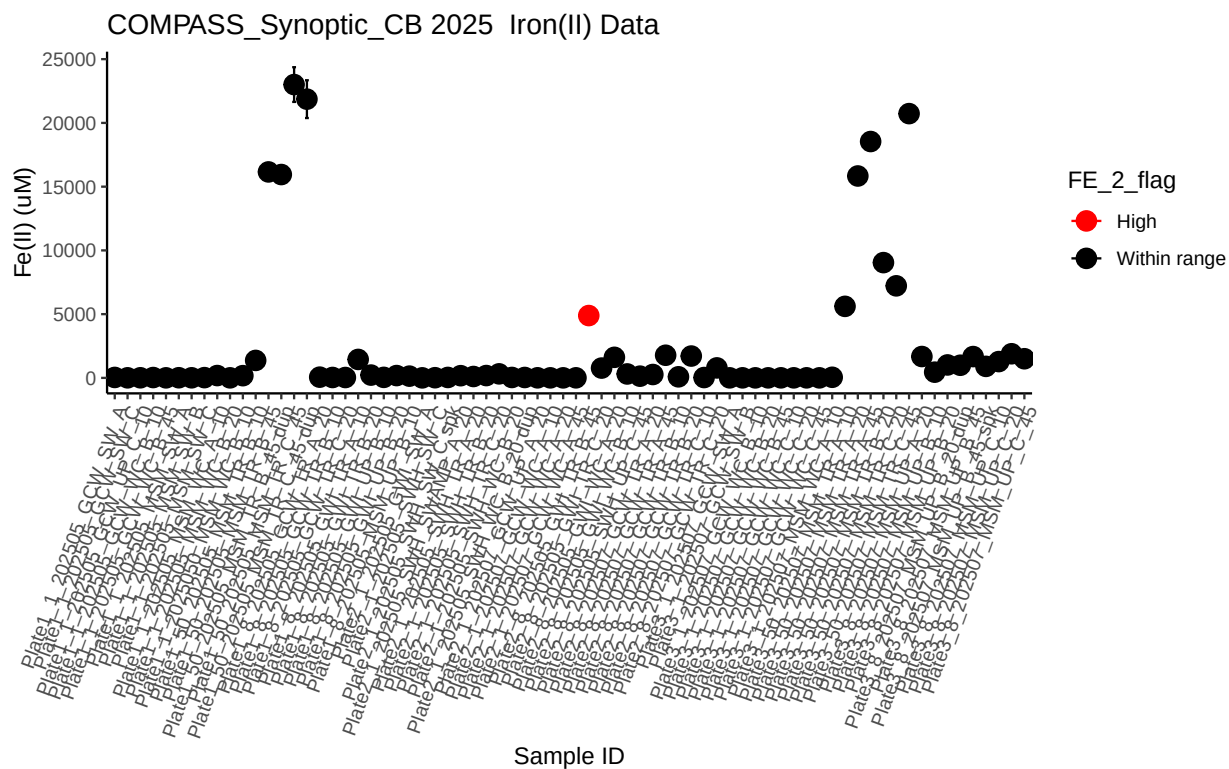
Table 4: Samples Above the Detection Limit that should be rerun at a higher dilution

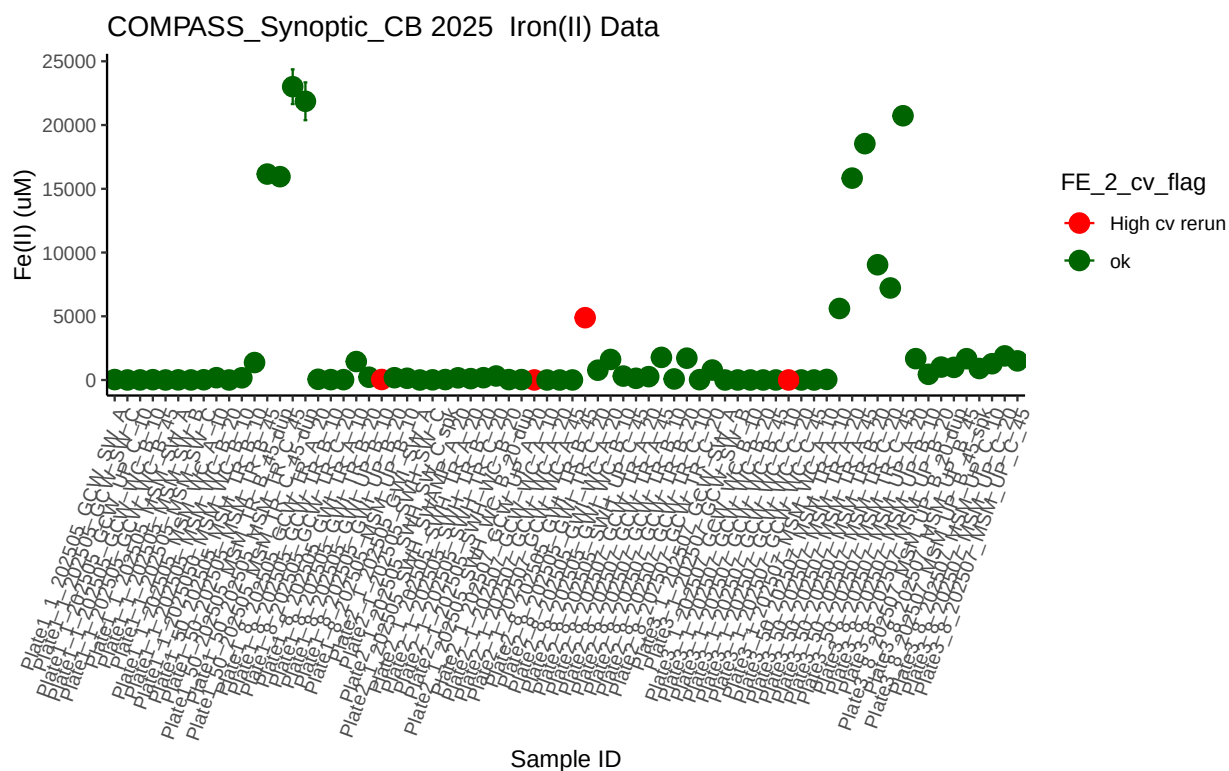
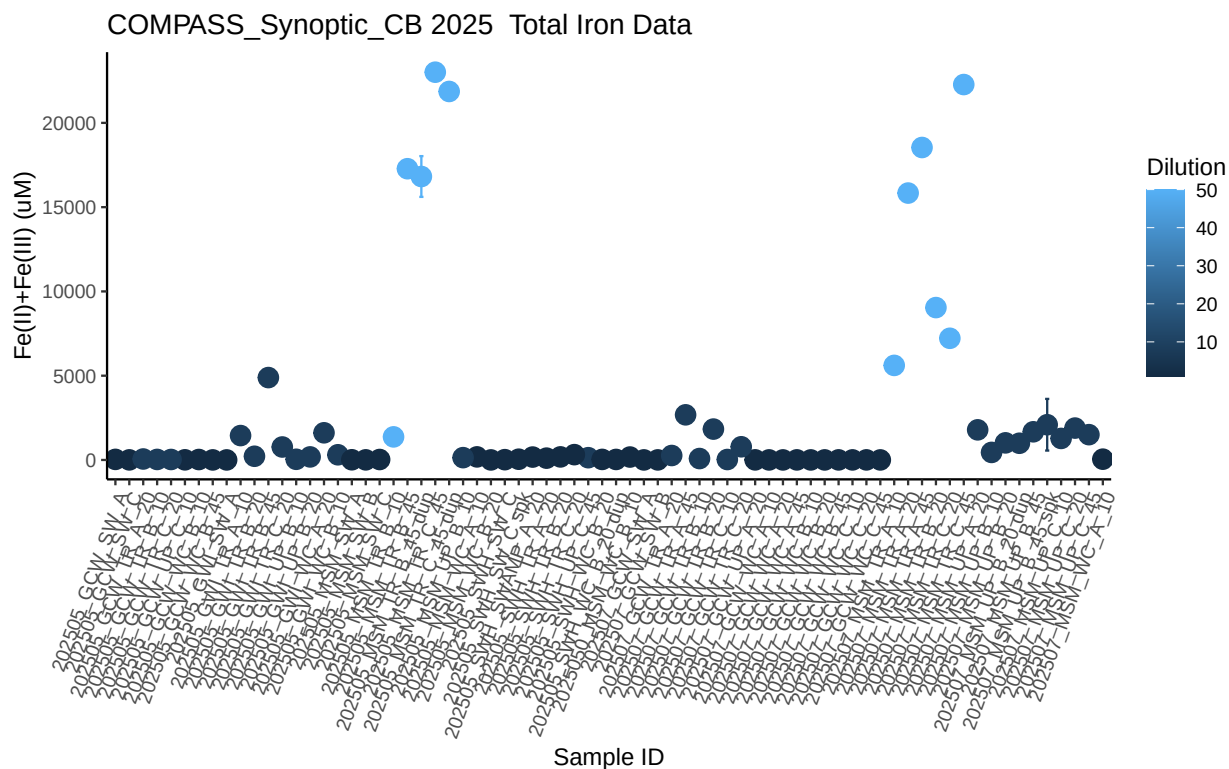
| ID full                      | FE 2 mean  | FE 3 mean | FE tot mean | FE 2 flag    | FE 3 flag |
|------------------------------|------------|-----------|-------------|--------------|-----------|
| Plate1_50_202505_MSM_TR_B_45 | 16154.6719 | 1125.3923 | 17280.0642  | Within range | High      |
| Plate2_8_202507_GCW_TR_A_45  | 1775.8717  | 902.2151  | 2678.0868   | Within range | High      |
| Plate2_8_202507_GCW_TR_B_20  | 1713.1479  | 119.3746  | 1832.5225   | Within range | High      |
| Plate3_50_202507_MSM_TR_C_45 | 20728.2833 | 1545.9276 | 22274.2108  | Within range | High      |
| Plate3_8_202507_MSM_UP_A_20  | 1675.2522  | 111.8195  | 1787.0717   | Within range | High      |
| Plate3_8_202507_MSM_UP_C_20  | 1892.1721  | 0.0000    | 1892.1721   | Within range | High      |
| Plate2_8_202505_GWI_TR_B_45  | 4888.5409  | 0.0000    | 4888.5409   | High         | High      |
| Plate3_8_202507_MSM_UP_C_45  | 1501.4550  | 0.0000    | 1501.4550   | Within range | High      |
| Plate1_1_2025050_MSM_WC_B_10 | 173.7977   | 0.0000    | 173.7977    | Within range | High      |
| Plate1_1_202505_MSM_WC_A_10  | 182.6182   | 0.0000    | 182.6182    | Within range | High      |
| Plate1_50_202505_MSM_TR_C_45 | 23006.9218 | 0.0000    | 23006.9218  | Within range | High      |
| Plate1_8_202505_GWI_TR_A_10  | 1447.8784  | 0.0000    | 1447.8784   | Within range | High      |
| Plate2_1_202505_SWH_TR_B_20  | 182.2915   | 0.0000    | 182.2915    | Within range | High      |
| Plate2_1_202505_SWH_TR_C_20  | 315.7430   | 0.0000    | 315.7430    | Within range | High      |
| Plate2_8_202505_GWI_WC_A_20  | 1612.5284  | 0.0000    | 1612.5284   | Within range | High      |
| Plate3_50_202507_MSM_TR_A_20 | 15836.1525 | 0.0000    | 15836.1525  | Within range | High      |
| Plate3_50_202507_MSM_TR_A_45 | 18539.4835 | 0.0000    | 18539.4835  | Within range | High      |
| Plate3_50_202507_MSM_TR_B_20 | 9041.0728  | 0.0000    | 9041.0728   | Within range | High      |
| Plate3_8_202507_MSM_UP_B_45  | 1671.3320  | 0.0000    | 1671.3320   | Within range | High      |

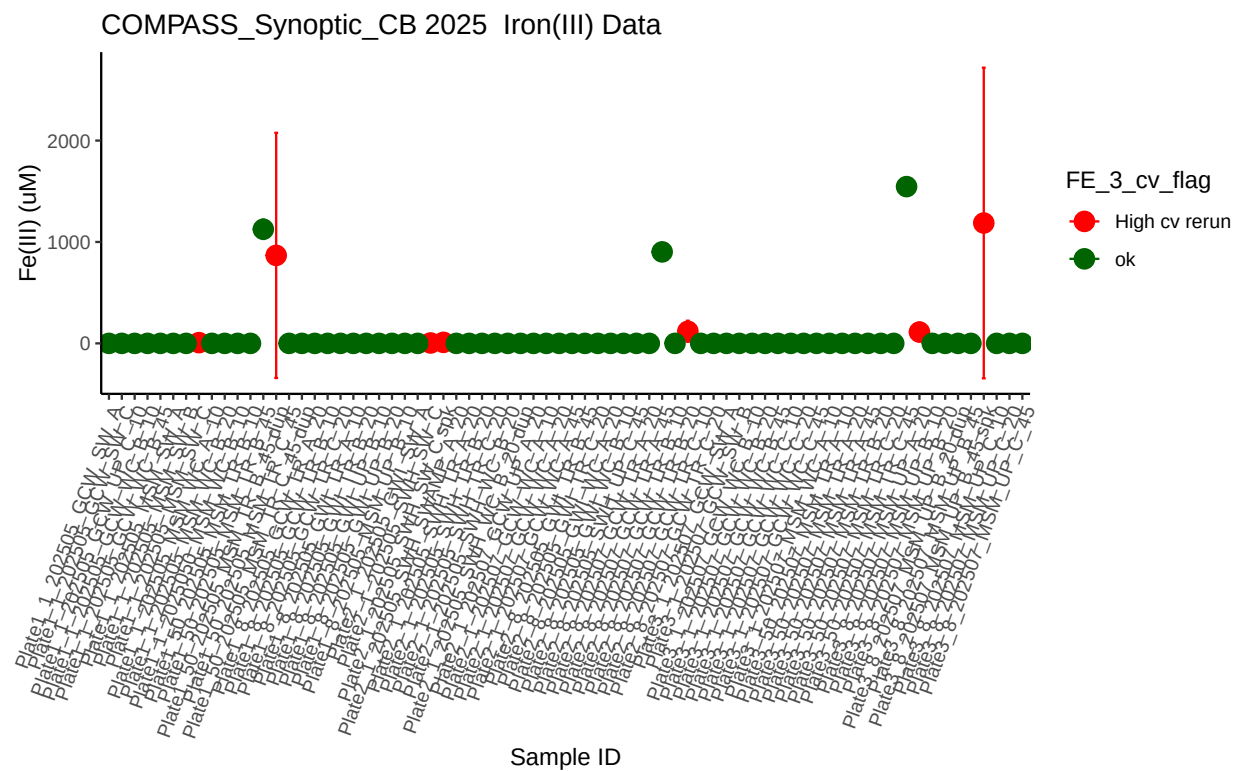
Table 5: Samples Below the Detection Limit that should be rerun at a lower dilution

| ID full                      | FE 2 mean  | FE 3 mean | FE tot mean | FE 2 flag    | FE 3 flag |
|------------------------------|------------|-----------|-------------|--------------|-----------|
| Plate1_8_202505_GCW_TR_C_20  | 28.58598   | 0         | 28.58598    | Within range | Below MDL |
| Plate1_8_202505_GWI_TR_B_20  | 220.22612  | 0         | 220.22612   | Within range | Below MDL |
| Plate1_8_202505_GWI_UP_B_10  | 40.75361   | 0         | 40.75361    | Within range | Below MDL |
| Plate2_8_202505_GWI_WC_B_10  | 308.44142  | 0         | 308.44142   | Within range | Below MDL |
| Plate2_8_202505_SWH_UP_C_45  | 135.05273  | 0         | 135.05273   | Within range | Below MDL |
| Plate2_8_202507_GCW_TR_C_10  | 27.06503   | 0         | 27.06503    | Within range | Below MDL |
| Plate3_8_202507_MSM_UP_B_10  | 448.36914  | 0         | 448.36914   | Within range | Below MDL |
| Plate3_8_202507_MSM_UP_C_10  | 1278.00140 | 0         | 1278.00140  | Within range | Below MDL |
| Plate1_50_202505_MSM_TR_B_10 | 1370.07595 | 0         | 1370.07595  | Within range | Below MDL |
| Plate1_8_202505_GCW_TR_A_20  | 57.99108   | 0         | 57.99108    | Within range | Below MDL |
| Plate1_8_202505_GCW_TR_B_10  | 31.62789   | 0         | 31.62789    | Within range | Below MDL |
| Plate1_8_202505_GWI_UP_B_20  | 181.69530  | 0         | 181.69530   | Within range | Below MDL |
| Plate1_8_202505_MSM_UP_B_10  | 138.09463  | 0         | 138.09463   | Within range | Below MDL |
| Plate2_8_202505_GWI_TR_C_20  | 769.67717  | 0         | 769.67717   | Within range | Below MDL |
| Plate2_8_202507_GCW_TR_A_20  | 264.84076  | 0         | 264.84076   | Within range | Below MDL |
| Plate2_8_202507_GCW_TR_B_10  | 84.35428   | 0         | 84.35428    | Within range | Below MDL |
| Plate2_8_202507_GCW_TR_C_20  | 782.74463  | 0         | 782.74463   | Within range | Below MDL |
| Plate3_50_202507_MSM_TR_A_10 | 5610.86427 | 0         | 5610.86427  | Within range | Below MDL |
| Plate3_50_202507_MSM_TR_C_20 | 7219.79541 | 0         | 7219.79541  | Within range | Below MDL |
| Plate3_8_202507_MSM_UP_B_20  | 1015.34543 | 0         | 1015.34543  | Within range | Below MDL |

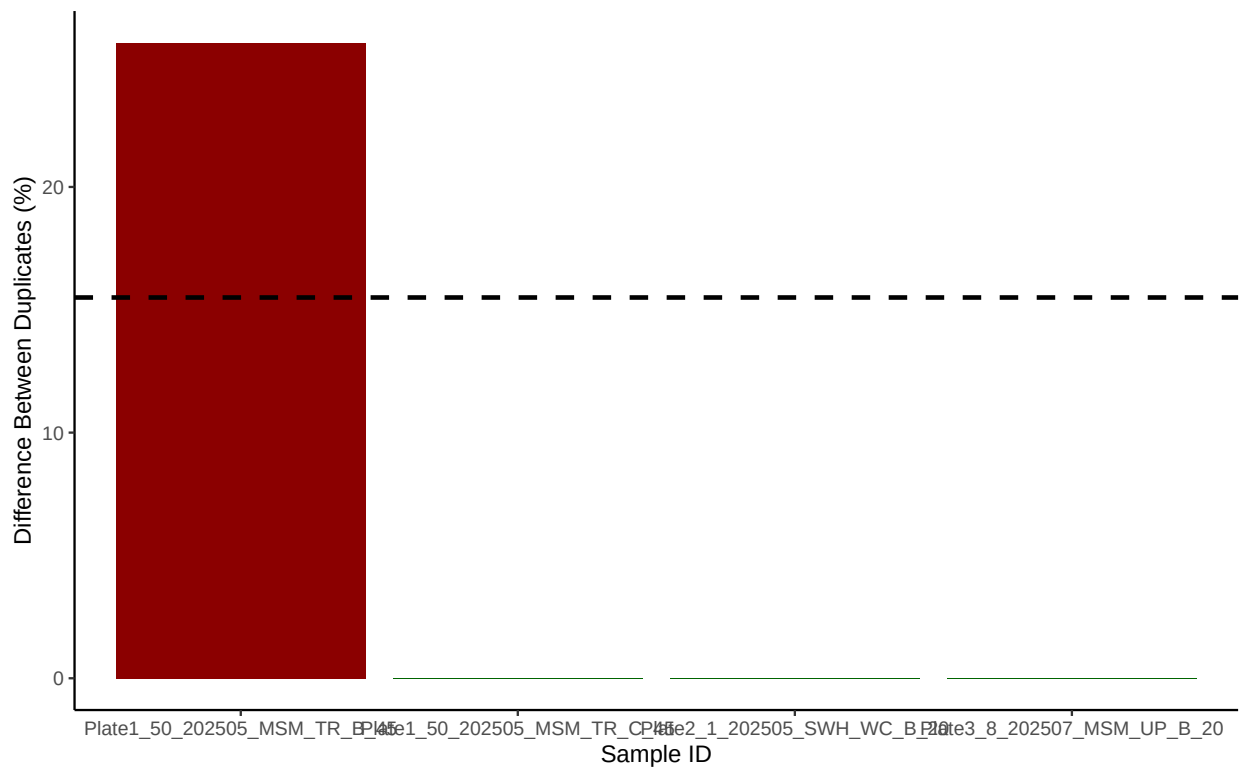
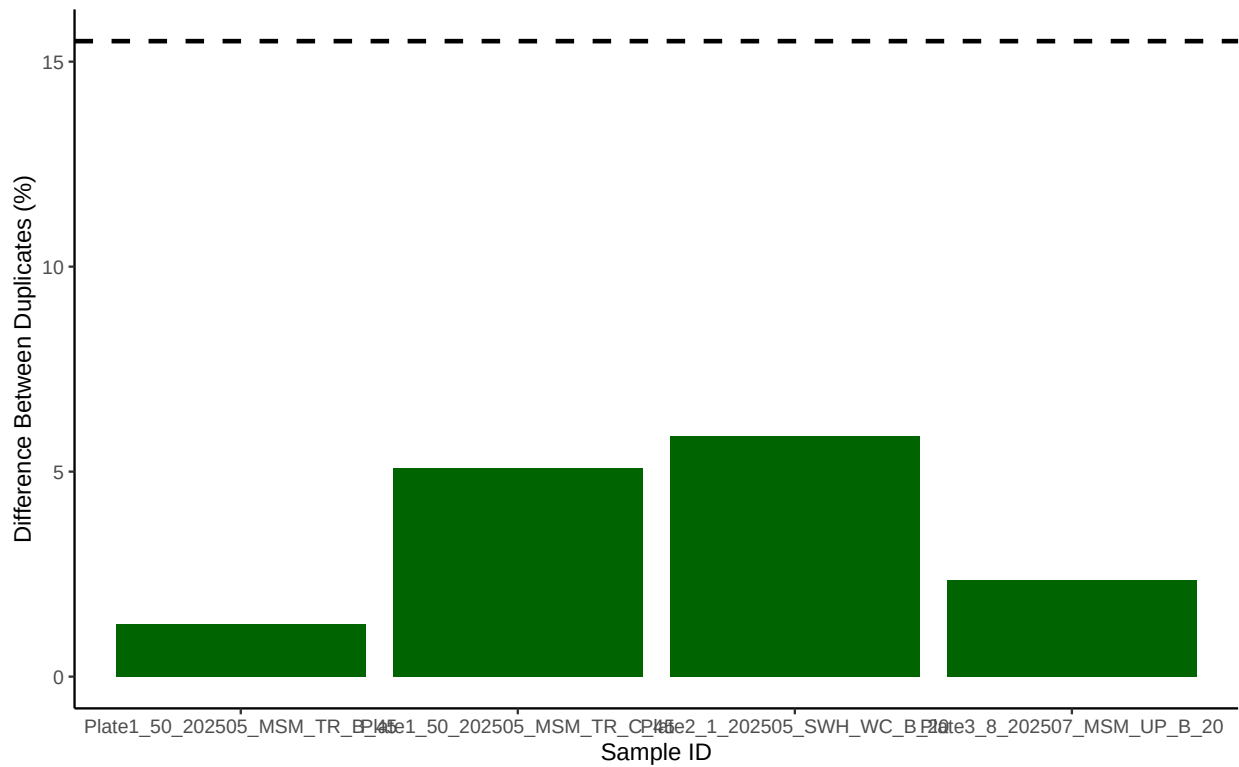
Plot data after bad reps were removed

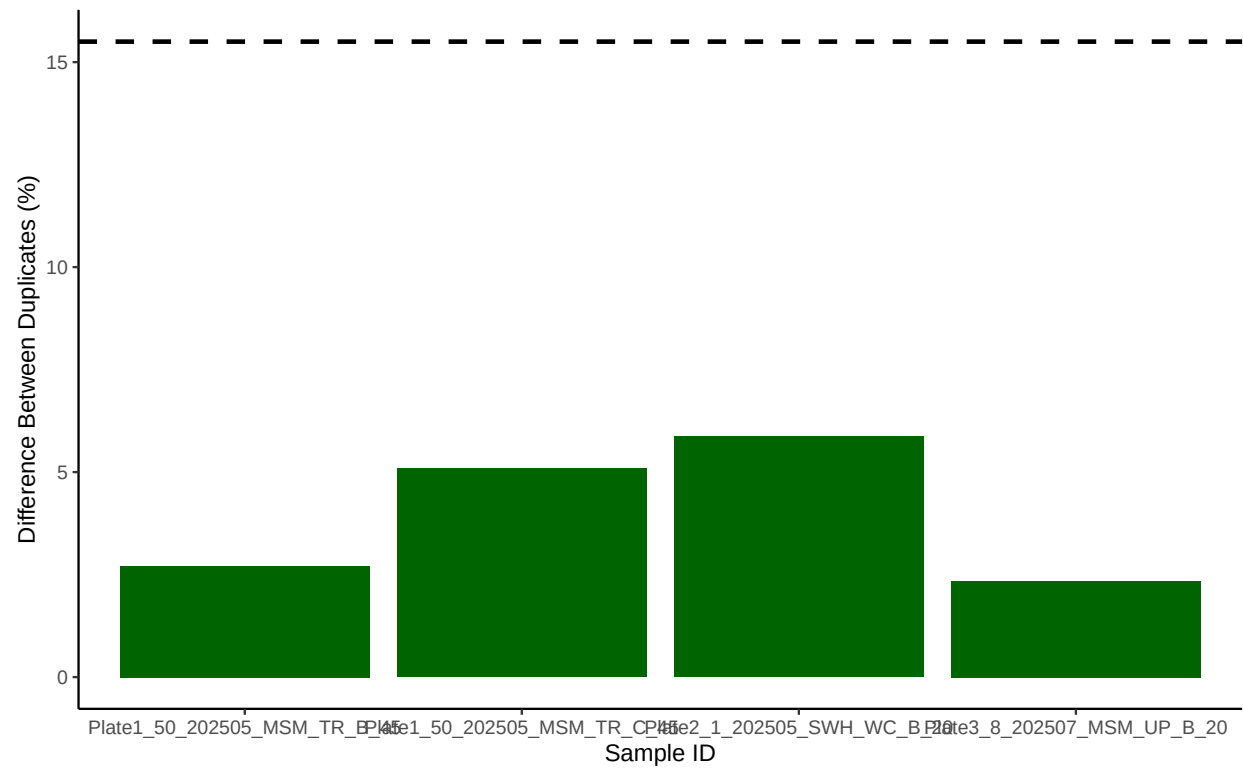






Check Dups for QAQC



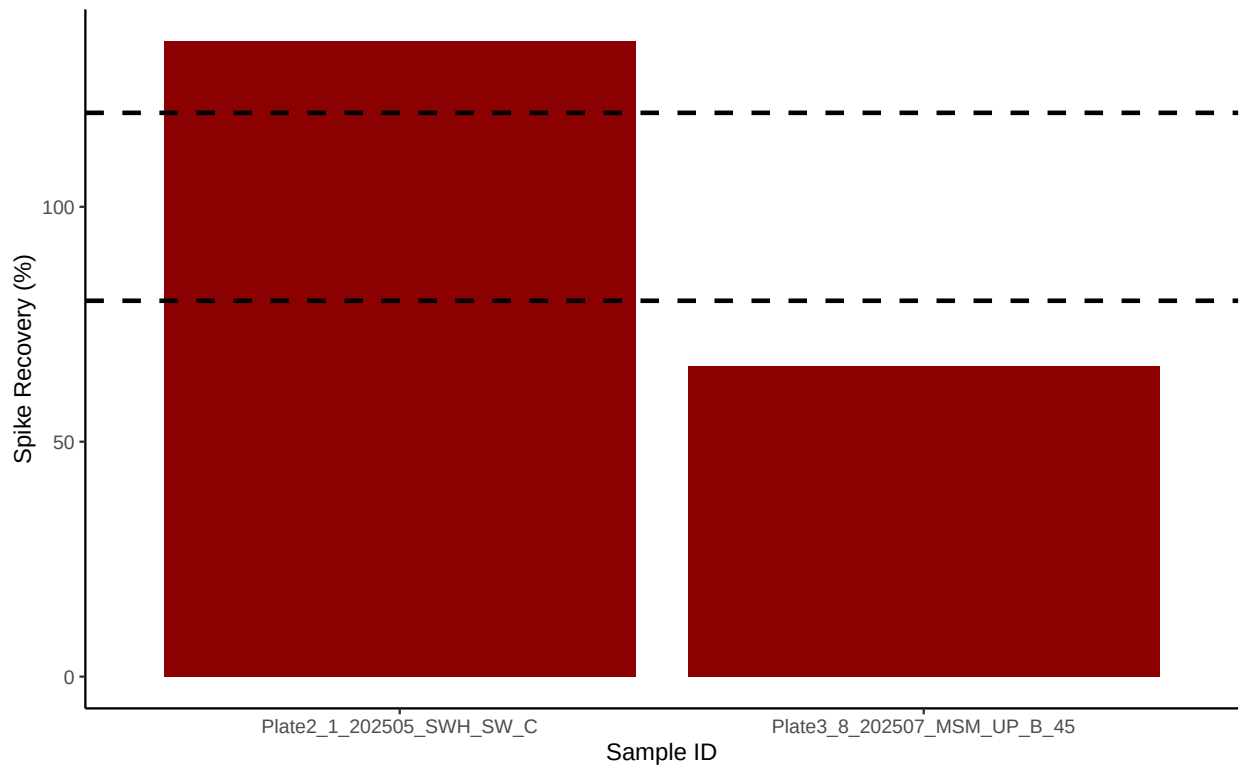


```
## [1] ">60% of Fe 2 Duplicates are within <10%"
```

```
## [1] ">60% of Fe 3 Duplicates are within <10%"
```



Check the spks for QAQC



```
## [1] "<60% of Spikes are out of range - REASSESS"
```

Export full data then just final data